

Stroke Assessment Tools – Full Guide with Scoring

1. NIH Stroke Scale (NIHSS) – Full Scoring Guide

Purpose: Quantifies severity of stroke symptoms.

0–42 points total. Higher score = more severe stroke.

1a. LOC (Alertness)

0 = Alert; 1 = Not alert, arousable; 2 = Requires repeated stimulation; 3 = Unresponsive

1b. LOC Questions (Month and Age)

0 = Both correct; 1 = One correct; 2 = Neither correct

1c. LOC Commands (Open/Close Eyes, Grip/Release Hands)

0 = Both correct; 1 = One correct; 2 = Neither

2. Best Gaze

0 = Normal; 1 = Partial gaze palsy; 2 = Forced deviation

3. Visual Fields

0 = No visual loss; 1 = Partial hemianopia; 2 = Complete hemianopia; 3 = Bilateral blindness

4. Facial Palsy

0 = Normal; 1 = Minor; 2 = Partial; 3 = Complete

5 & 6. Motor Arm/Leg (R and L)

0 = No drift; 1 = Drift; 2 = Some effort against gravity; 3 = No effort; 4 = No movement

7. Limb Ataxia

0 = Absent; 1 = Present in one limb; 2 = Present in two

8. Sensory

0 = Normal; 1 = Mild/moderate; 2 = Severe

9. Best Language

0 = No aphasia; 1 = Mild/moderate; 2 = Severe; 3 = Mute

10. Dysarthria

0 = Normal; 1 = Mild/moderate; 2 = Severe

11. Extinction/Inattention (Neglect)
0 = No neglect; 1 = Mild; 2 = Severe

2. Cincinnati Prehospital Stroke Scale (CPSS)

Purpose: Fast stroke screen in EMS.

1. Facial Droop:

Ask patient to smile or show teeth

- Normal: Both sides move equally
- Abnormal: One side does not move

2. Arm Drift:

Close eyes, hold both arms out straight for 10 seconds

- Normal: Both arms move equally or not at all
- Abnormal: One arm drifts down or cannot lift

3. Speech:

Ask patient to repeat a simple phrase

- Normal: Correct and clear
- Abnormal: Slurred, incorrect, or unable

3. Los Angeles Prehospital Stroke Screen (LAPSS)

Purpose: EMS stroke triage tool.

Inclusion Criteria:

- Age >45
- No history of seizures
- New symptoms <24h
- Was ambulatory before event
- BG between 60–400

Exam:

- Unilateral facial droop
- Grip strength weakness
- Arm drift or weakness

If all criteria are met + ≥ 1 exam abnormality → Stroke suspected.

4. RACE Scale – Rapid Arterial Occlusion Evaluation

Purpose: LVO detection for EMS

Domains (0–2 or 0–3 scale):

- Facial palsy (0–2)
- Arm motor (0–2)
- Leg motor (0–2)
- Head/gaze deviation (0–1)
- Aphasia (dominant hemisphere, 0–2)
- Agnosia (nondominant hemisphere, 0–2)

Score ≥ 5 suggests LVO

5. VAN – Vision, Aphasia, Neglect

Purpose: Fast bedside LVO screen

Step 1: Arm Weakness?

- If no → VAN negative
- If yes → Check:
 - Visual disturbance
 - Aphasia (speech difficulty)
 - Neglect (ignoring one side)

If arm weakness + ≥ 1 of above → VAN positive = High LVO risk

6. BE-FAST

Public-friendly stroke awareness mnemonic

- B = Balance (sudden loss of coordination)
- E = Eyes (vision loss/double vision)
- F = Face (facial droop)
- A = Arm (weakness or drift)
- S = Speech (slurred or strange)
- T = Time (note time, call 911)

7. MED Score

Purpose: Emerging LVO screen tool

- Motor deficit (arm or leg): 1 point
- Eye deviation: 1 point
- Denial or neglect: 1 point

Score ≥ 2 = Suggestive of LVO

8. Modified Rankin Scale (mRS)

Purpose: Functional disability scoring after stroke

- 0 = No symptoms
- 1 = No significant disability; can carry out all usual duties
- 2 = Slight disability; unable to do all previous activities
- 3 = Moderate disability; needs some help
- 4 = Moderately severe; needs assistance for bodily needs
- 5 = Severe disability; bedridden
- 6 = Death

References

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4. **RACE Scale**

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5. **VAN (Vision, Aphasia, Neglect) Screening Tool**

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VAN: A bedside tool to identify large vessel occlusion strokes. *J Stroke Cerebrovasc Dis*. 2016;25(4):984–989.

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6. **BE-FAST Mnemonic**

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<https://www.stroke.org/en/about-stroke/stroke-symptoms>

7. **MED Score**

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8. **Modified Rankin Scale (mRS)**

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